

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078

Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“App Development and ML Model Training Project for Foot Pressure Analyzer Device”

Submitted in partial fulfillment for the award of degree(20INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

Hitha N - 1DS21AI023

Conducted at



**SPV4 Ameya Innovation Labs Private
Limited**

**Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78**

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CERTIFICATE

This is to certify that the Internship titled “**App Development and ML Model Training Project for Foot Analyzer Device**” carried out by **Ms. HITHA N [1DS21AI023]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering, in Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

Signature of Reporting Head
Dr. Sandip Kumar Roy
Director
SPV4 Ameya

Signature of Guide
Dr. Aruna M G
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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DECLARATION

I, **Hitha N**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **SPV4 Ameya Innovation Labs Private Limited**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS21AI023

Place: Bangalore

NAME: Hitha N

OFFER LETTER



SP4 AMEYA INNOVATION LABS PRIVATE LIMITED



OFFER LETTER OF INTERNSHIP

Date: 07/09/2024

Sub: Internship Offer - App Development and ML Model Training Project

Dear Hitha N,

We are pleased to offer you the internship position for the "App Development and ML Model Training Project" associated with our Foot Pressure Analyzer Device. Your skills and interests align well with the project requirements, and we are excited to have you on board.

We look forward to your acceptance and a fruitful collaboration.

Yours Faithfully,

A handwritten signature in blue ink, appearing to read "Sandip", followed by a small blue dot.

SANDIP KUMAR ROY
(Director)
DIN: 10532220

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COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. I take this opportunity to express my gratitude to all who have helped me to complete the Internship.

I express my sincere thanks to the Principal **Dr. B G Prasad**, for providing me adequate facilities to undertake this Internship.

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I would like to thank **Dr. Sandip Kumar Roy**, Director, SPV4 Ameya Innovation Labs Private Limited for guiding me during the period of internship.

I express my deep and profound gratitude to my Guide, **Dr. Aruna M G**, Associate Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

I would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

I would like to thank the non-teaching members of the Department of AI & ML, for helping me during the Internship.

Last but not the least, I would like to thank my parents and friends without whose constant help, the completion of Internship would have not been possible.

HITHA N [1DS21AI023]

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CHAPTER 1

INTRODUCTION

INTRODUCTION TO FOOT PRESSURE ANALYZER DEVICE AND APP

Foot pressure analyzer device is an essential diagnostic tool used by healthcare professionals to evaluate foot function, gait patterns, and balance. By measuring the distribution of pressure across different areas of the foot during standing or walking, this method helps in detecting abnormalities, monitoring rehabilitation progress, and preventing injuries. It is particularly valuable for individuals with conditions like diabetes, arthritis, or those undergoing physiotherapy and post-surgical recovery.

To bridge the gap between clinical assessment and everyday monitoring, Safe Step, a smart mobile application that has been developed as a digital solution for real-time foot pressure analysis. The app is designed to work in conjunction with wearable foot pressure sensors and provides users with detailed insights into their step count, balance, and overall walking pattern.

Safe Step is particularly tailored for patients and doctors, offering personalized analytics, progress tracking, and communication features. By integrating Machine Learning (ML) algorithms, the app can detect deviations from normal walking patterns and alert users or caregivers to potential risks.

This internship project at SP4 Ameya Innovation Labs Private Limited focused on the development and enhancement of the Safe Step app, involving key tasks like Flutter-based mobile development, Firebase integration, and training ML models to interpret sensor data and generate meaningful feedback.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

SPV4 AMEYA INNOVATION LABS PRIVATE LIMITED



ABOUT THE COMAPNY

SP4 Ameya Innovation Labs Private Limited is a healthcare technology company focused on transforming the MedTech landscape through the development of smart, accessible, and efficient healthcare solutions. Established with the objective of bridging the gap between digital health transformation and patient-centric care, the organization specializes in creating scalable and affordable physiotherapy devices, wearable rehabilitation tools, and AI-integrated healthcare applications.

The company integrates advanced technologies such as Artificial Intelligence (AI), the Internet of Things (IoT), and cloud computing to offer solutions that address both current and future challenges in the healthcare ecosystem. With a strong emphasis on inclusivity, SP4 Ameya Innovation Labs designs its products and services to improve accessibility for individuals across all levels of physical ability and to enhance the operational efficiency of healthcare providers.

Innovations developed at SP4 Ameya are targeted towards improving therapeutic outcomes, increasing patient safety, and delivering real-time, data-driven insights for healthcare decision-making. Through collaborative research and development, the company continues to redefine standards in smart medical technology and rehabilitation science.

VISION:

To bridge the gap between healthcare needs and digital transformation by enabling accessible, intelligent, and personalized healthcare innovations that enhance quality of life and deliver patient-centric care.

MISSION:

To pioneer accessible healthcare innovations by leveraging AI-powered analytics, modular product designs, and IoT-enabled devices. The mission encompasses enhancing therapeutic efficiency, improving safety standards, and delivering scalable, affordable, and portable

solutions compatible with telehealth platforms. The company focuses on providing technology-driven tools that support predictive healthcare delivery and empower patients and providers alike.

CORE AREAS OF INNOVATION:

- **Wearable Rehabilitation Tools:**

Design and development of ergonomic, data-driven wearable devices that support physiotherapy and rehabilitation. These tools enable real-time monitoring of patient movements and therapy effectiveness, facilitating precise treatment planning.

- **AI-Integrated Applications:**

Development of mobile and web-based applications powered by artificial intelligence to support automated diagnosis, progress tracking, and personalized treatment recommendations. These applications are designed to improve engagement and outcomes for both patients and healthcare professionals.

- **IoT and Cloud-Based Infrastructure:**

Implementation of IoT-enabled devices connected through cloud platforms to ensure seamless data collection, synchronization, and access across multiple systems. This infrastructure supports real-time analytics and enhances the efficiency of healthcare delivery.

- **Modular and Portable Medical Devices:**

Creation of compact, customizable, and transportable devices suitable for use in clinical settings as well as at-home care environments. These devices are built to be cost-effective and scalable, ensuring accessibility across a wide user base.

- **Telehealth-Compatible Solutions**

Integration of telemedicine features into products and applications to facilitate remote healthcare delivery. This includes virtual consultations, remote monitoring, and digital health reporting, ensuring continuous care even in remote or underserved locations.

CHAPTER 3

SCOPE OF WORK

The scope of work undertaken during the internship at SP4 Ameya Innovation Labs Private Limited focused on active contribution toward the design, development, and testing phases of healthcare mobile applications. The primary project involved was the Safe Step App, a smart foot pressure monitoring system integrated with machine learning and cloud-based architecture. The internship experience also included testing responsibilities for a previously developed application by the organization, thereby providing well-rounded exposure to the software development life cycle (SDLC), quality assurance, and user experience design in a MedTech environment.

1. Application Testing (Initial Phase)

The internship began with involvement in the quality assurance and testing phase for one of the existing applications developed by the company. This experience provided foundational knowledge of the product lifecycle and enabled a deeper understanding of:

- Application flow and feature functionality
- Bug identification and reporting
- UI/UX evaluation and usability assessment
- Test case execution and feedback documentation

The testing tasks included both functional and non-functional evaluation of the application on Android devices. Bugs and performance issues were reported using structured formats, and suggestions for improvement were communicated to the development team. This phase served as a valuable entry point into the domain of MedTech applications and helped establish the testing standards followed by the organization.

2. Wireframe Contribution for Safe Step App

One of the significant contributions was toward the UI/UX wireframing of the Safe Step App, which is a mobile application designed to analyze foot pressure data collected from wearable sensors. The wireframe design was created using Figma, and efforts were directed toward:

- Designing intuitive and accessible user interfaces tailored for doctors and patients.
- Structuring app navigation flow to support ease of use and clarity in user interaction.

- Ensuring that the app reflected the company's vision of inclusivity, efficiency, and patient-centered care
- Providing design feedback and collaborating with developers for UI refinement

The wireframe played a critical role in setting the foundation for the final app design. Attention was given to accessibility, minimalism, and a clean layout to ensure a seamless user experience, especially for patients with limited digital literacy.

3. Frontend Development Using Flutter

Post wireframe design, the internship involved hands-on contribution to frontend development using the Flutter framework, which is known for its cross-platform capabilities and fast development cycle. Key tasks included:

- Assisting in the implementation of the splash screen and navigation logic based on Firebase authentication status
- Developing UI components and integrating interactive elements as per the wireframe
- Customizing buttons, input fields, and layout structures in alignment with the design vision
- Debugging interface issues and enhancing responsiveness across screen sizes

The frontend phase provided exposure to Flutter widgets, state management, routing, and platform-specific adaptations, thereby strengthening mobile development skills in a real-world project.

4. Backend Integration Using Firebase

Alongside frontend development, work was carried out on backend integration using Firebase, which served as the cloud-based backend for authentication, data storage, and user management. The following tasks were undertaken:

- Implementing user authentication using FirebaseAuth
- Configuring Firestore database structure for storing patient and therapist data
- Setting up data fetch and write operations to connect the frontend with the backend
- Participating in the setup of user navigation logic based on authentication and profile completion flags

This integration ensured secure and scalable cloud-based operations for the Safe Step app, aligning with SP4 Ameya's commitment to IoT-ready, telehealth-compatible healthcare solutions.

The internship involved diverse tasks spanning app testing, UI/UX design, mobile frontend development, and cloud backend integration, enabling comprehensive involvement in the app development process. By contributing to both technical and design aspects of the Safe Step app, and engaging in the testing of a prior project, the internship offered hands-on exposure to real-world software development in the context of smart healthcare applications.

CHAPTER 4

LEARNING OBJECTIVES

1. Gaining Practical Exposure to Mobile App Development Using Flutter:

One of the primary learning objectives was to gain practical, hands-on experience in mobile app development using the Flutter framework. Flutter, being a modern and widely-used cross-platform development toolkit, enabled the creation of highly responsive and visually appealing user interfaces. During the internship, tasks involved implementing various UI components, managing screen navigation, customizing widgets, and ensuring consistent behavior across different devices. Working on the Safe Step app provided an opportunity to learn how real-world mobile applications are structured, including how design layouts, routing, and state management are handled in Flutter. This objective helped develop a deep understanding of app architecture and the best practices for efficient and scalable mobile development.

2. Understanding Backend Integration and Cloud Services with Firebase:

Another key objective was to understand the integration of cloud-based backend services using Firebase. Firebase, a comprehensive development platform provided by Google, was used in the Safe Step app for user authentication, data storage, and real-time synchronization. The internship offered hands-on experience with configuring FirebaseAuth for managing login and sign-up processes, integrating Firestore to store and retrieve user data, and implementing conditional navigation logic based on profile completion flags. This learning helped solidify concepts such as asynchronous operations, cloud data management, and backend security — all of which are critical for building modern mobile applications that are scalable and user-friendly.

3. Designing User-Centric Wireframes and UI Structures:

Contributing to the UI/UX wireframe design was a significant part of the internship experience, aimed at understanding the principles of user-centric design. Using tools like Figma, efforts were made to translate functional requirements into interactive wireframes that formed the foundation of the Safe Step app. This objective involved learning about user flow mapping, interface hierarchy, consistency, and accessibility—especially important in healthcare applications where users may include elderly individuals or patients with limited mobility. The process provided insight into how design decisions impact usability, and how to collaborate effectively with developers to bring those wireframes to life while maintaining the intended user experience.

4. Developing Testing and Debugging Skills in a Real-World Setting:

The internship also focused on developing effective software testing and debugging skills. Initially, responsibilities included serving as a tester for another application created by the company. This experience offered exposure to real-world testing procedures including test case execution, bug tracking, performance evaluation, and user feedback collection. Understanding the importance of structured testing methodologies helped in identifying functionality gaps and UI/UX inconsistencies. Later, during the development of the Safe Step app, these skills were applied to debug UI behavior and resolve issues in data handling, navigation, and backend communication. This objective reinforced the importance of iterative development and quality assurance in professional app development.

5. Exploring the Role of Technology in Healthcare and Rehabilitation:

A distinctive objective of this internship was to explore how modern technologies like AI, IoT, and mobile platforms are transforming the healthcare and rehabilitation landscape. Working on the Safe Step app, which is built to analyze foot pressure data for gait assessment and physiotherapy monitoring, provided insights into the domain-specific challenges of MedTech development. The internship allowed observation of how data from wearable sensors can be used to drive clinical insights and how mobile platforms can make therapy more accessible and effective. This learning bridged the gap between theoretical knowledge and practical application, especially in understanding how digital health solutions can empower both patients and healthcare providers.

CHAPTER 5

CHALLENGES AND SOULTION

1. Managing Conditional Navigation Based on User Profile:

Challenges:

The Safe Step app required a dynamic navigation system that could direct users to different screens depending on their authentication status and whether their profile and personal details were completed. Managing this logic within the splash screen using FirebaseAuth and Firestore proved to be tricky initially. Errors such as delayed redirection, incorrect route handling, or null data fetching interrupted the flow and sometimes led to app crashes or unintended behavior.

Solution:

To resolve this, Firebase initialization and asynchronous data fetching were optimized within the initState() method of the splash screen. Proper null checks and default values were introduced to avoid unhandled exceptions. A clearly structured if-else condition tree was created that relied on boolean flags like profileCompleted and detailsCompleted, which were accurately retrieved from Firestore before deciding the navigation path. Debugging with log prints helped trace the issue flow, and a delay was added using Future.delayed to give Firebase enough time to initialize properly. This approach streamlined the navigation logic and ensured a smooth user experience during app launch.

2. Syncing Real-Time Data with Firebase Firestore:

Challenges:

A major challenge was ensuring seamless real-time syncing of user data between the app and Firestore, especially during updates such as completing user profiles, tracking steps, or modifying health parameters. There were instances where the UI did not reflect the updated data instantly, or duplicate entries were generated due to improper stream handling or race conditions.

Solution:

This issue was addressed by adopting Firebase's real-time listeners using StreamBuilder, which ensured that any changes made to Firestore documents were immediately reflected in the UI. The Firestore rules were reviewed to ensure proper write and read permissions, and debouncing logic was implemented to prevent multiple rapid writes to the same document. Additionally, transaction-based updates were utilized for critical data entries to maintain data integrity. These steps collectively improved the reliability and accuracy of real-time data communication between the app and the backend.

3. Designing User-Friendly Wireframes for a MedTech App:

Challenges:

Designing a healthcare-focused app like Safe Step brought unique UI/UX challenges. Since the app targets users undergoing physiotherapy, some of whom may have mobility limitations or limited familiarity with mobile apps, it was critical to design an interface that is simple, intuitive, and accessible. Balancing aesthetics with functionality while keeping it aligned with medical standards was initially difficult, especially during the early stages of the wireframing process.

Solution:

To tackle this, the design process started with understanding the user personas and expected use cases. Inspiration was taken from existing healthcare and fitness apps to ensure familiarity. Wireframes were created using Figma with a strong focus on color contrast, font readability, and large, easily tappable buttons. Feedback was continuously gathered from team members and potential users to iterate on the designs. Dropdowns, icons, and onboarding hints were added to improve accessibility. The iterative design-thinking approach helped in creating wireframes that were both user-friendly and medically appropriate.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- Flutter SDK
- Firebase
- Android Studio
- Visual Studio Code
- Figma
- Git & GitHub
- Firebase CLI (Command Line Interface)
- Google Chrome / Android Emulator

Hardware Requirements:

- Processor: Intel i5 or more
- RAM: 8 GB minimum
- Display: 23 LED Display
- 64bit operating systems
- Android Smartphone
- LAN Connection Port: Required

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

1. App Testing & QA (Initial Phase):

During the initial phase of the internship, comprehensive manual testing was conducted on an existing in-house mobile application developed by SP4 Ameya Innovation Labs for a separate healthcare use-case. This involved systematically identifying and documenting various issues related to UI glitches, inconsistencies in the navigation flow, and challenges with responsiveness across different screen sizes and device types. Detailed observations were compiled into bug reports, which were shared with the development team for resolution. In addition to functional testing, constructive feedback was provided to enhance the overall user experience. Specific suggestions were made to improve the app's onboarding process, with the goal of simplifying initial setup and ensuring a smoother experience for first-time users. These contributions not only improved the quality and usability of the application but also fostered a deeper understanding of user-centric testing approaches and the importance of continuous refinement in digital health solutions.

2. Wireframe Design for Safe Step:

Active participation was made in brainstorming sessions focused on the UI/UX design of the Safe Step application, where creative inputs were shared to shape an intuitive and user-friendly interface. Collaboration using Figma played a crucial role in developing the wireframe structure of the app, which involved planning the overall navigation flow, strategic button placements, tab layouts, and onboarding screens tailored for both patients and doctors. A strong emphasis was placed on accessibility, ease of use, and clinical relevance—particularly for users with mobility challenges who require a straightforward and supportive digital experience. During the design process, enhancements were proposed and implemented for key sections such as the home screen dashboard, which displays step count and gait analysis insights, as well as for the login and sign-up flow to streamline user onboarding. Additionally, attention was given to the doctor interface, ensuring that it offered clear access to patient data, assessment reports, and communication features in a well-organized and efficient layout.



Fig 6.1: Splash Screen of Safe Step App

3. Front-End Development using Flutter:

Following the design phase, active involvement continued in transforming the wireframe designs into functional user interface screens using Flutter. This included working with a variety of Flutter widgets to bring the app's visual structure to life while maintaining consistency with the proposed layout. Efforts were made to ensure that the application followed responsive design principles, allowing it to render effectively across different screen sizes and orientations. Key contributions included supporting the creation of the splash screen, which featured the app's logo and incorporated FirebaseAuth initialization logic to determine user login status. Assistance was also provided in building the login and signup pages, complete with input validation and a streamlined user flow for seamless account creation. Navigation logic was implemented to enable conditional routing, allowing the app to dynamically redirect users based on their profile completion status. In addition, active debugging of UI issues and optimization of the codebase helped ensure clean, modular development practices, including the use of reusable widgets for better maintainability and scalability.

4. Firebase Integration:

Significant contributions were made to the backend integration of the Safe Step application by working extensively with Firebase services. Firebase Authentication was implemented to manage secure user login and signup functionality using email and password credentials, ensuring data protection and user account integrity. Cloud Firestore was utilized to store and retrieve vital user data such as personal profile information, steps taken, and other health-related metrics. To enhance user experience with live updates, real-time database listeners were integrated, enabling automatic synchronization between backend data changes and the app's user

interface. Additionally, collaboration with senior developers facilitated the development of an efficient navigation wrapper that dynamically directed users based on their Firebase authentication and profile completion status. This seamless backend integration significantly contributed to making the app responsive, secure, and tailored to individual user journeys.

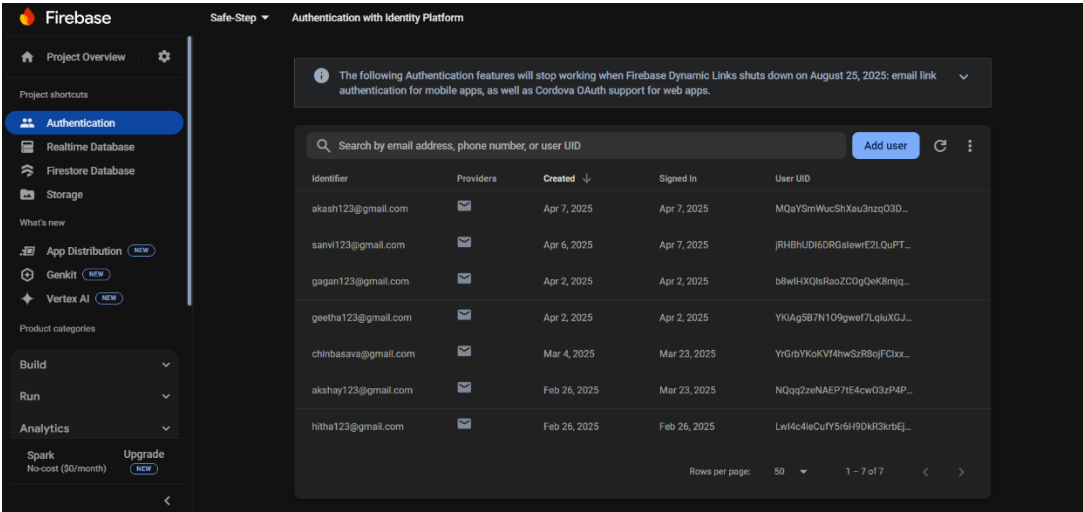


Fig 6.2: Firebase console depicting the authentication details of users

CHAPTER 7

REFLECTION AND ANALYSIS

The internship at SP4 Ameya Innovation Labs Pvt. Ltd. has been a highly enriching experience that provided a well-rounded exposure to the process of app development and machine learning model integration in the healthcare domain. Working on the *Safe Step* application not only helped build technical proficiency but also offered a deeper understanding of how digital tools can be thoughtfully designed and deployed to address real-world medical needs.

One of the most valuable aspects of this internship was the opportunity to contribute to various phases of the app development lifecycle—from conceptual design using wireframes to actual frontend and backend implementation using Flutter and Firebase. This multifaceted involvement enabled the development of a strong sense of ownership over the product and allowed practical learning across areas such as UI/UX design, user authentication, database management, and responsive development.

The collaborative work environment at SP4 Ameya encouraged open communication, which fostered creativity and helped refine problem-solving skills. Brainstorming sessions with peers and seniors not only enhanced design thinking but also taught the importance of empathy in healthcare tech—especially in making applications accessible and intuitive for users with mobility challenges.

Moreover, the challenges encountered during implementation—such as ensuring real-time data sync, managing screen responsiveness, and handling conditional navigation—served as important learning curves. These issues reinforced the need for writing modular, scalable code and emphasized the importance of thorough testing.

From a personal development perspective, this internship instilled confidence in working with professional tools like Figma, Flutter, and Firebase. It also strengthened soft skills like communication, adaptability, and analytical thinking, which are essential in both tech-driven and interdisciplinary team environments.

CHAPTER 8

RECOMMENDATIONS

- 1. Enhance Cross-Platform Testing:** To ensure the *Safe Step* app offers a consistent user experience across a variety of devices, it is recommended to conduct regular cross-platform testing on different screen sizes and operating systems. This will help identify UI inconsistencies and performance issues early in the development cycle.
- 2. Implement Role-Based Access Control (RBAC):** While the app currently supports patient and doctor profiles, adding clearly defined role-based access control will further strengthen data privacy and improve user management. Doctors should have controlled access to patient health data, while patients should have autonomy over their personal information.
- 3. Integrate In-App Feedback System:** Adding a built-in feedback module will allow users to report bugs or suggest improvements directly within the app. This real-time feedback can be invaluable for continuous improvement and building user trust.
- 4. Improve Onboarding Experience:** Though functional, the onboarding flow can be further simplified and made more engaging using step-by-step tooltips, animations, or short tutorials. This would ease the learning curve, especially for elderly or less tech-savvy users.
- 5. Enhance Real-Time Analytics Dashboard:** Developing a real-time dashboard for healthcare professionals would allow them to visualize trends in patient activity and gait data. This can support predictive diagnosis and improve clinical decision-making.

CHAPTER 9

CONCLUSION

The internship at SP4 Ameya Innovation Labs Pvt. Ltd. has been an invaluable and transformative experience that offered deep insights into the real-world application of app development and machine learning within the healthcare domain. The opportunity to work on the *Safe Step* application not only allowed practical exposure to industry-standard tools such as Flutter, Firebase, and Figma, but also highlighted the importance of user-centric design in developing digital health solutions.

Contributing to various aspects of the project—from initial testing of an existing healthcare app, to brainstorming UI/UX ideas, designing wireframes, implementing frontend features, and integrating Firebase backend services—provided a well-rounded experience. Each phase helped in building technical depth, problem-solving abilities, and collaboration skills, which are critical for success in any tech-driven role.

Through this internship, a greater understanding was developed of how medical technology must prioritize accessibility, usability, and data security. It became clear that thoughtful design and robust backend infrastructure are key to creating applications that can truly benefit users, especially those with physical limitations or chronic conditions.

Moreover, working in a startup-like environment fostered agility and adaptability, encouraging active learning and innovation. The guidance from mentors and collaboration with experienced developers helped in bridging the gap between academic learning and practical application, especially in areas like UI/UX principles, real-time data sync, and user authentication logic.

Overall, this internship has not only strengthened foundational skills in app development and backend integration but has also sparked a deeper interest in the med-tech sector and its potential to revolutionize healthcare delivery. The experience has reaffirmed a commitment to pursuing a career that blends technology, empathy, and innovation for meaningful impact in people's lives.

CHAPTER 10

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